

efficiency gain from this, going from tasks that take hours to tasks that take seconds.

Can you provide examples of some difficult problems solved by your algorithm?

One problem with battery design is that of a combinatorial explosion of possibilities due to the number of different cell models, configurations, and other parts.

Think of it like chess. The first chess computers quickly outperformed humans because they could analyse hundreds of thousands of moves quickly. The way people design battery packs now is like chess—humans use knowledge, experience, and intuition to constrain the total search size. This must lead to less than optimal results since the full search space of potential battery designs has not been created or analysed.

One of the ways we solved this problem is with a highly optimised algorithm that can generate the designs thousands of times per second. A second part is how we analyse and sort the results from most to least useful to present users with the optimal design.

What benefit can battery design engineers receive compared to any other similar product? What is your USP?

Other battery software we have seen focus on simulations, especially for EV packs. We focus on automating the design of any battery pack so that users can start manufacturing as soon as possible.

How is that different from other battery design tools?

From what we have seen, other battery design software are usually a sub-feature of their main software. They have high-cost software for enterprises that are designing electric vehicles and large-format battery packs. These tools focus on simulations, particularly thermal simulations.

Think of a company like Toyota designing a new electric vehicle. Our software is for any business that designs battery packs, everyone ranging from DIY to startups to enterprises. It is lower-cost than competitors, and we are more focused on small and medium format packs, although we can produce large format packs as well. That means robots, drones, home power systems, or anything else you can imagine powering is what can benefit from our software.

Another big difference is our vision and overall focus on automation and Industry 4.0. The tools we provide are intended to speed up the engineering workflow—both to find the optimal design quickly and output the necessary information to begin manufacturing. What we are building today will run tomorrow's smart battery factories while our competitor products are not guided by this vision.

Which companies and institutions are using your software? What kind of products are they developing?

There is a wide range of people using the software. We have seen one company build a battery design for a giant solar yacht. We have seen electric bikes, robots, drones, home solar systems, and many other applications. **EFY**

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